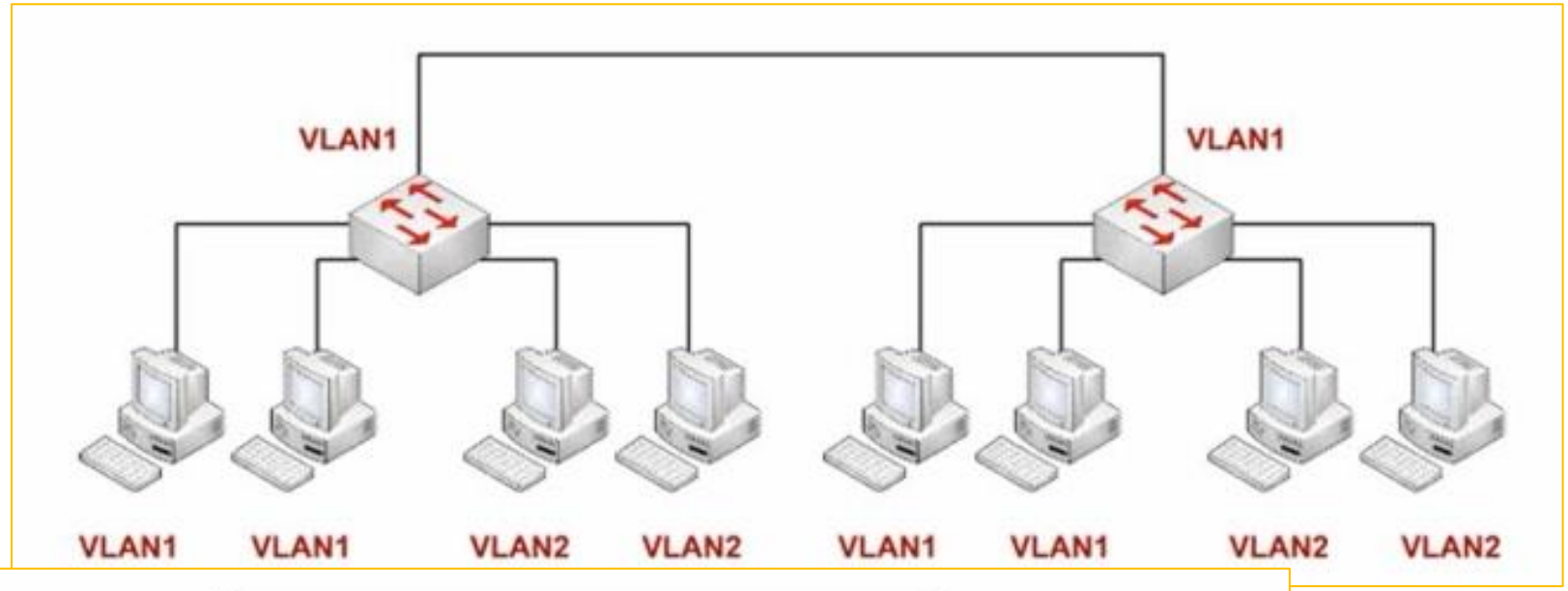
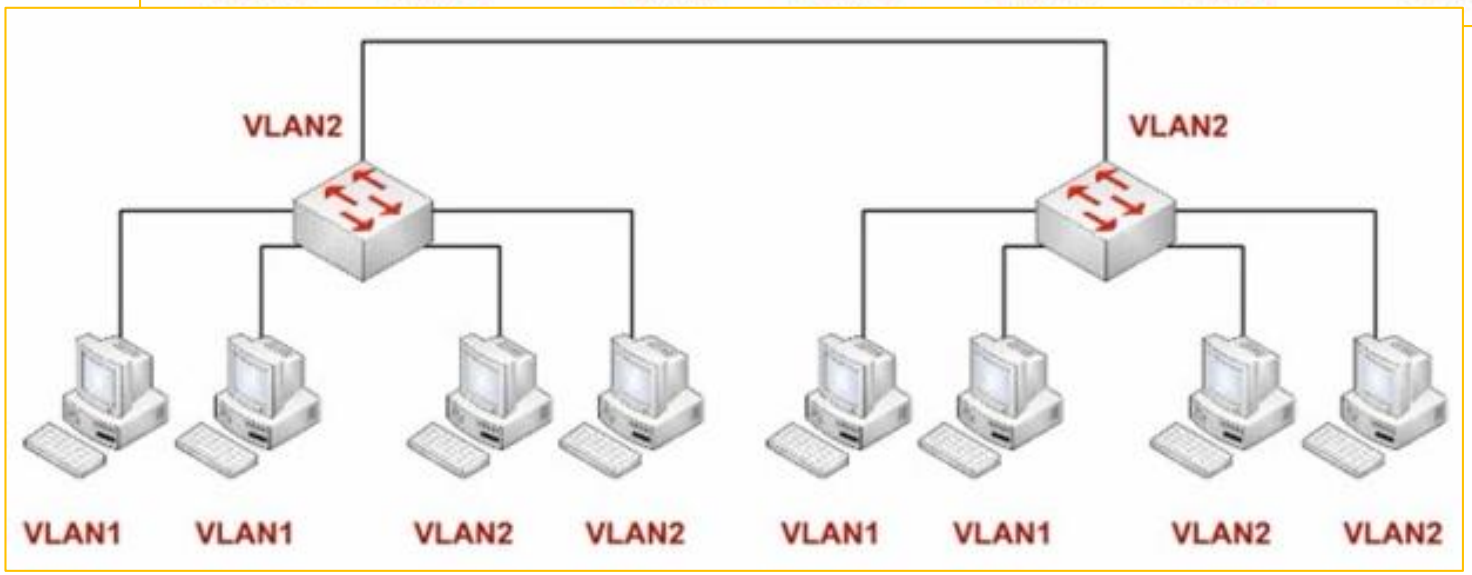


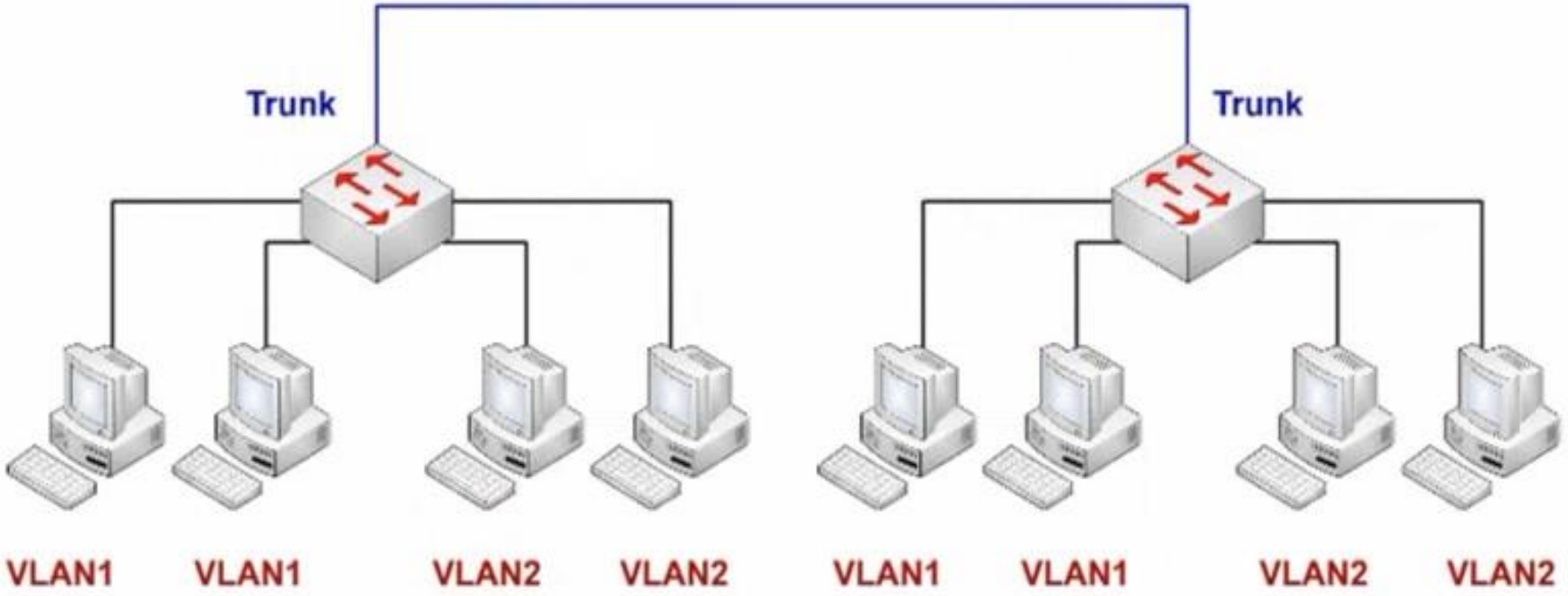


VLAN Trunks

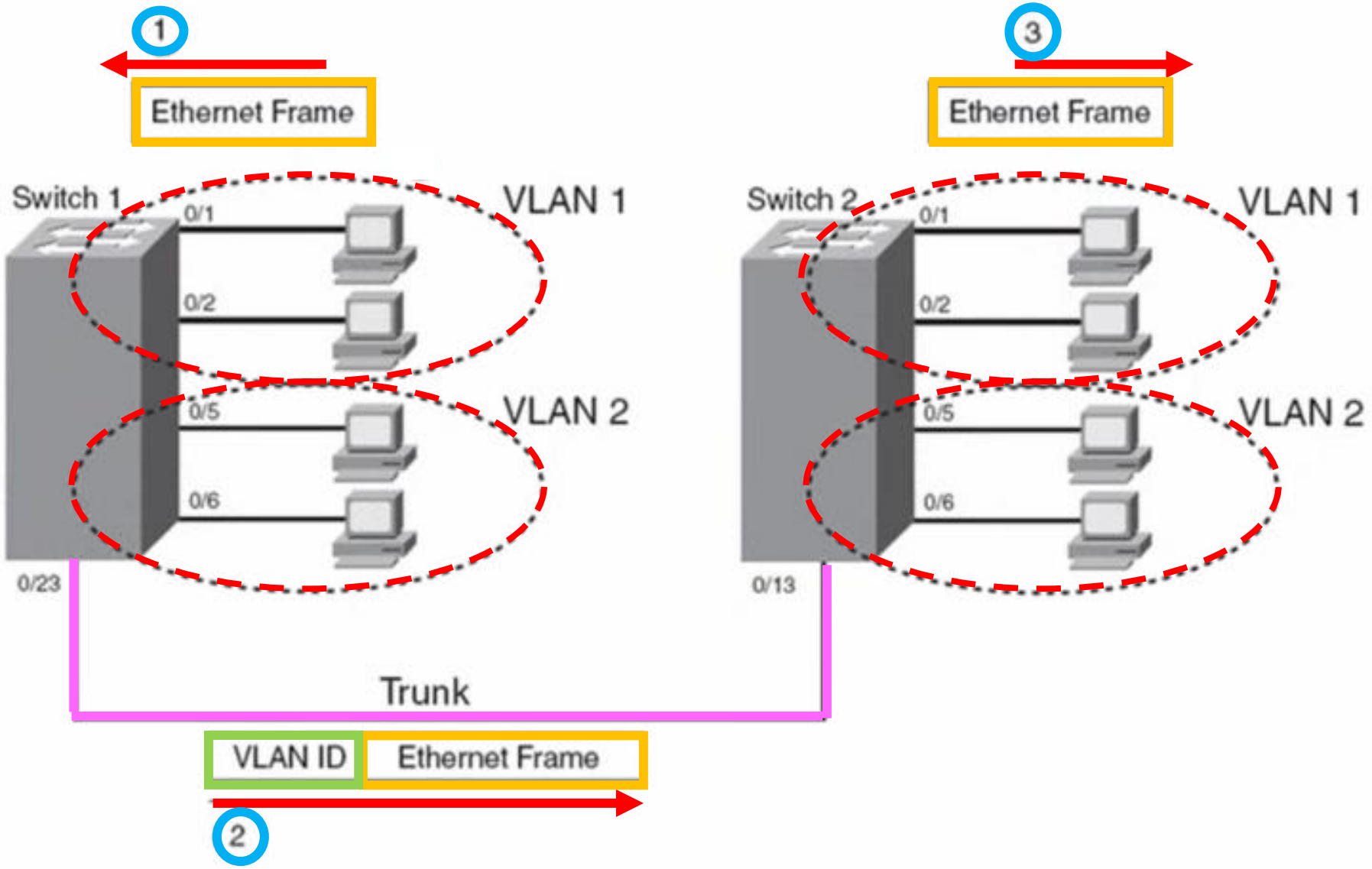

VLAN1
VLAN1
VLAN1
VLAN1
VLAN2
VLAN2
VLAN1
VLAN1
VLAN2
VLAN2


VLAN2
VLAN2
VLAN1
VLAN1
VLAN2
VLAN2
VLAN1
VLAN1
VLAN2
VLAN2

VLAN Trunks Switching

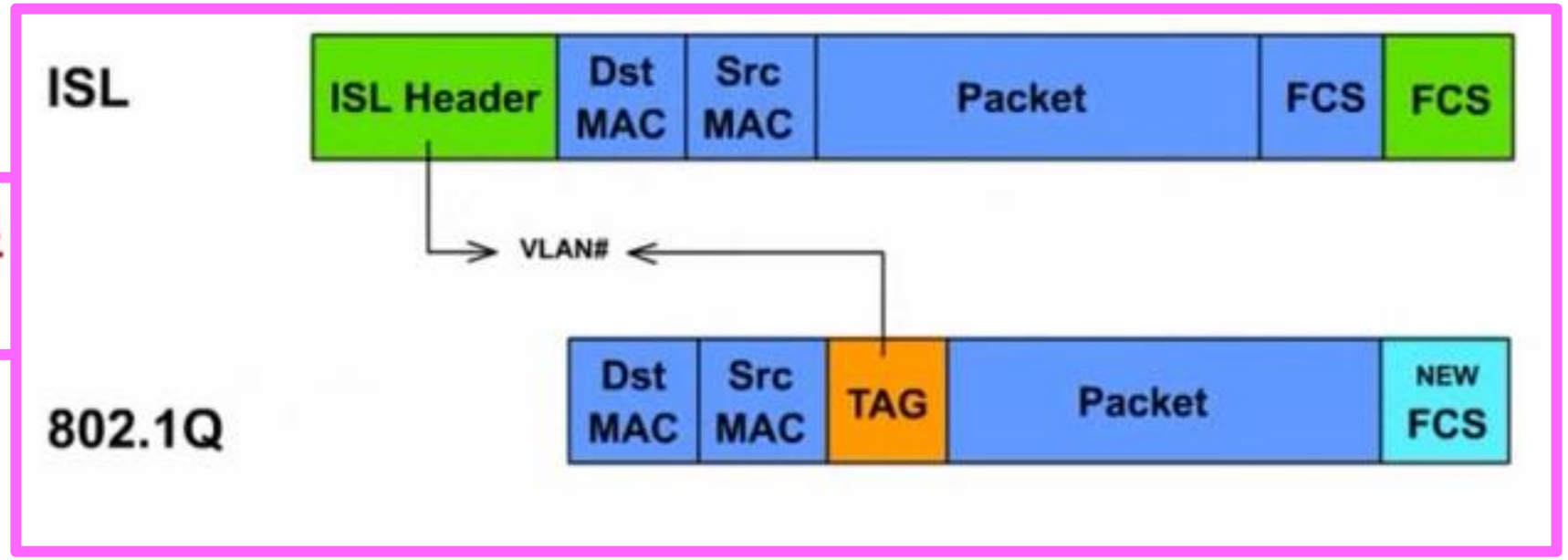


VLAN Trunks Switching



Trunking protocols

ISL Vs 802.1Q



2. 802.1Q is an IEEE standard while ISL is Cisco
3. *Native VLAN* is only supported in 802.1Q



```
Switch(config)# interface fastethernet <slot/port>
Switch(config-if)# switchport trunk encapsulation {802.1Q|ISL}
Switch(config-if)# switchport mode trunk
```

Example

```
Switch(config)# interface fastethernet 0/10
Switch(config-if)# switchport trunk encapsulation 802.1Q
Switch(config-if)# switchport mode trunk
```

Configuration

Switch(config-if)#switchport trunk allowed vlan *<vlan-id>,<vlan-id>,…*
Switch(config-if)#switchport trunk allowed vlan {remove|add} *<vlan-id>*
Switch(config-if)#switchport trunk native *<vlan-id>*

Example

Switch(config-if)#switchport trunk allowed vlan *1,2,3*
Switch(config-if)#switchport trunk allowed vlan add *4*
Switch(config-if)#switchport trunk allowed vlan remove *3*
Switch(config-if)#switchport trunk native *2*

Command Option	Description
access	Prevents the use of trunking, making the port always act as an access (nontrunk) port
trunk	Always uses trunking
dynamic desirable	Initiates negotiation messages and responds to negotiation messages to dynamically choose whether to start using trunking and defines the trunking encapsulation
dynamic auto	Passively waits to receive trunk negotiation messages, at which point the switch will respond and negotiate whether to use trunking, and if so, the type of trunking



Dynamic Trunking Protocol (DTP)

Expected Trunking Operational Mode Based on the Configured Administrative Modes

Administrative Mode	Access	Dynamic Auto	Trunk	Dynamic Desirable
access	Access	Access	Access	Access
dynamic auto	Access	Access	Trunk	Trunk
trunk	Access	Trunk	Trunk	Trunk
dynamic desirable	Access	Trunk	Trunk	Trunk



Dynamic Trunking Protocol (DTP)

Configuration

Switch(config-if)# Switchport mode dynamic {desirable | auto}

Disabling DTP on trunks

Switch(config-if)# Switchport mode Trunk
Switch(config-if)# Switchport nonegotiate